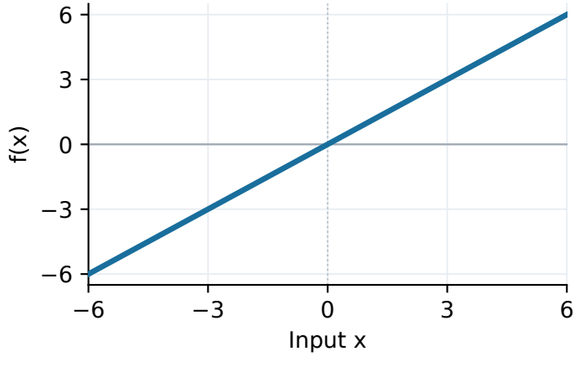
Common ANN activation functions

Same input range in every panel. Output scales vary so that bounded and negative responses stay visible.

PReLU slope is learned (one example shown). Swish uses beta = 1. Softmax is a vector-valued function.

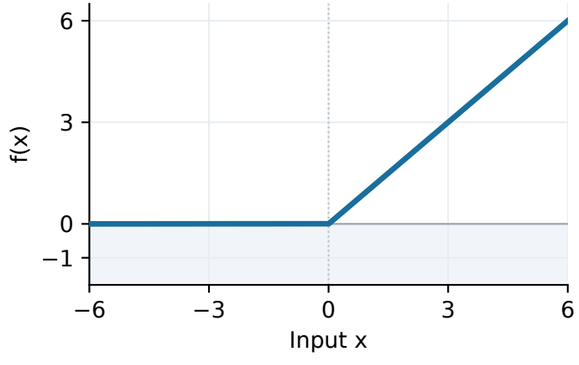
A Linear

$f(x) = x$



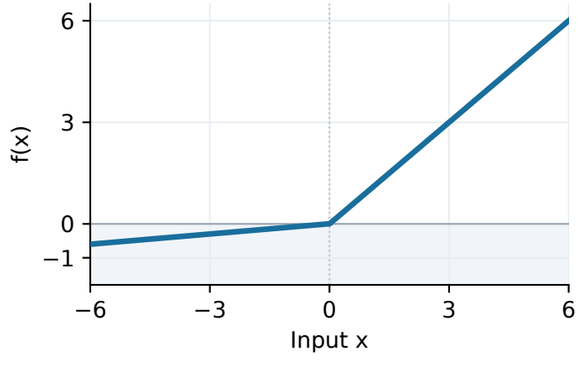
B ReLU

$f(x) = \max(0, x)$



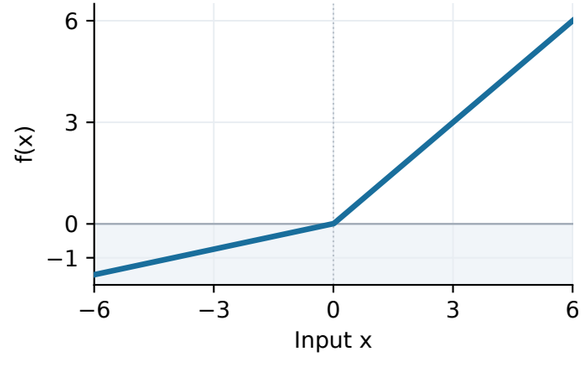
C Leaky ReLU

$\max(0, x) + 0.1\min(0, x)$



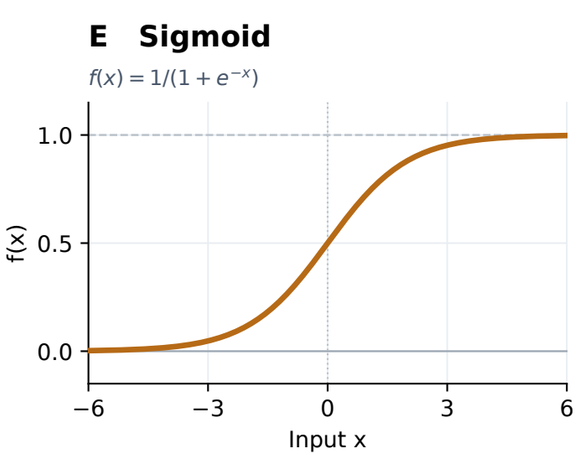
D PReLU

$\max(0, x) + a\min(0, x)$; example $a = 0.25$



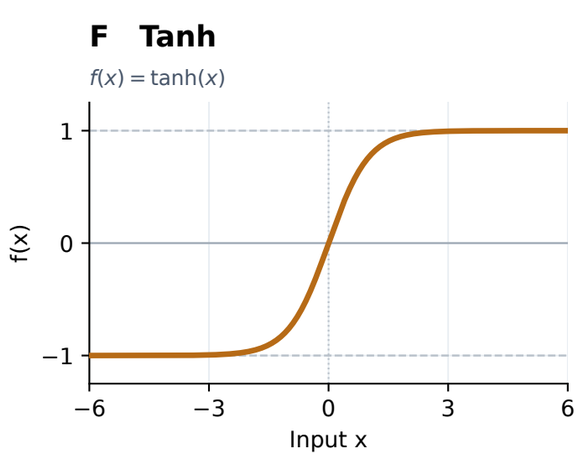
E Sigmoid

$f(x) = 1/(1 + e^{-x})$



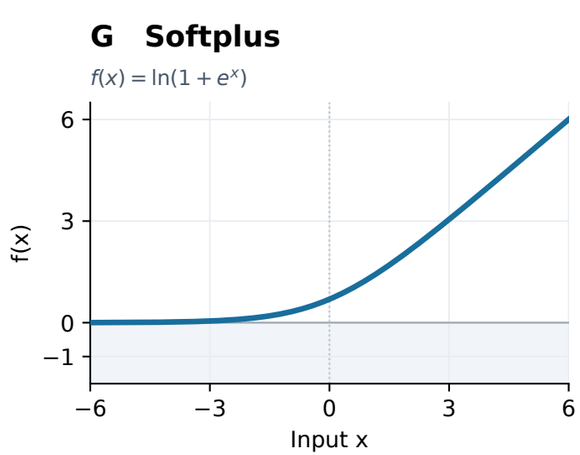
F Tanh

$f(x) = \tanh(x)$



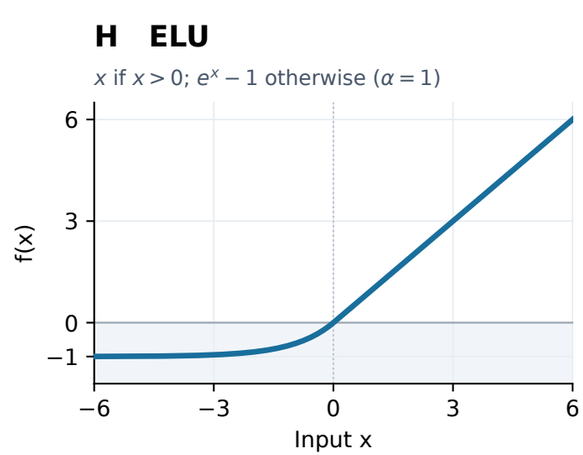
G Softplus

$f(x) = \ln(1 + e^{x})$



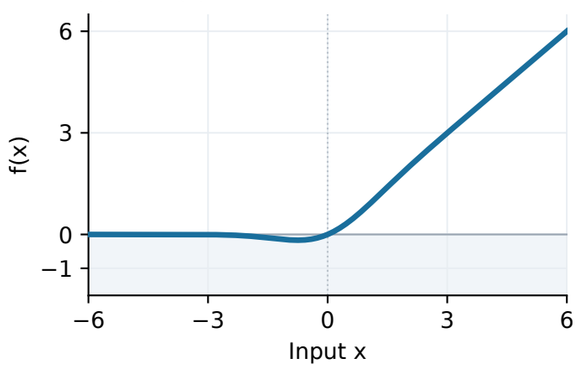
H ELU

x if $x > 0$; $e^{x} - 1$ otherwise ($\alpha = 1$)



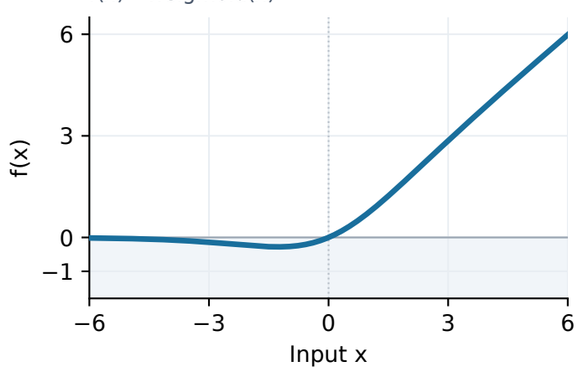
I GELU

$f(x) = x\Phi(x)$



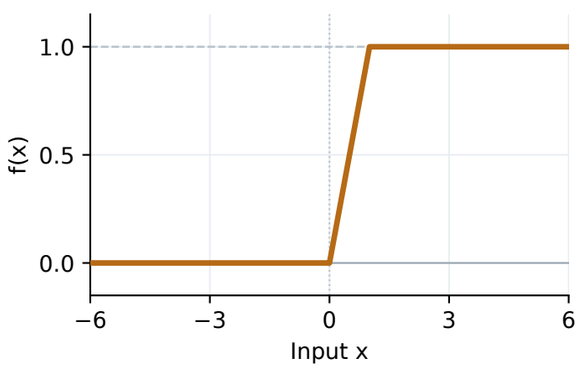
J SiLU / Swish

$f(x) = x\,\mathrm{sigmoid}(x)$



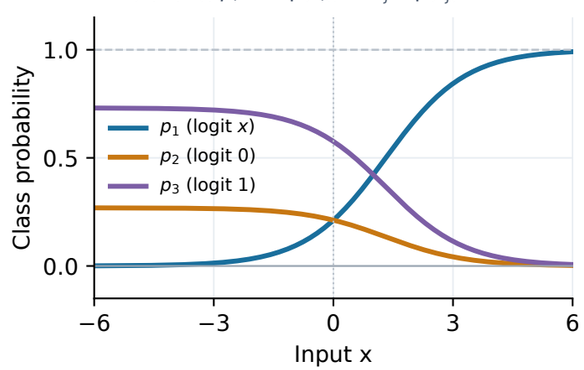
K Clipped ReLU

$f(x) = \min(1, \max(0, x))$



L Softmax

$\mathbf{z} = (x, 0, 1)$; $p_i = \exp(z_i)/\mathrm{sum}_j \exp(z_j)$



Orange single curves have an upper bound. GELU uses the exact Gaussian CDF; no tanh approximation.